Github Project Report

Group members

Jacob Ford - JacobPaulFord
Brendan Valleau - Brendanv55
Cayden Jones - Cool22j
Jack Miller - J-Mill3
Marc Edwards - heronic-studios

Repository link

https://github.com/Team-2-IGP/Homework2

Game description

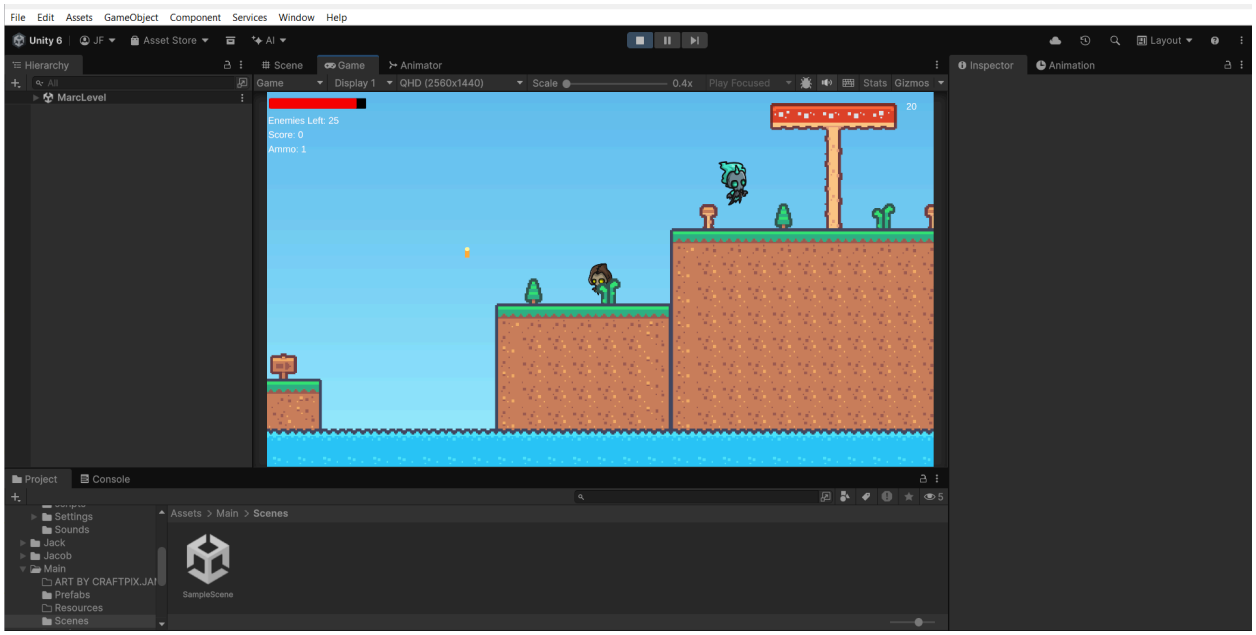
- Main objective: Kill all the enemies
- A and D to move left and right respectively
- Space to jump
    - You can jump on enemies to kill them
- Shift to shoot in your current direction

Contribution table and Commit Evidence

Name	Contributions	Each Major Commit
Jacob Ford	1. Basic back-and-forth movement enemy 2. Fall zone death trigger 3. Healing Collectible	1. Created basic enemy and added enemy and wall layers for interactions 2. Create Fall Zone Prefab close #14 3. Added healing object updating (#5)
Brendan Valleau	1. Tilemap & Art 2. Level creation 3. Animations	1. Tilemap and Art 2. Level designed 3. Animation and map update

Jack Miller	1. Movement bug fixed where the player would stop horizontal movement if coming in contact with a wall while airborne, even though they were holding a direction. 2. Added an ammo collectible that updates the UI when collected 3. Allows the player to shoot horizontally (by pressing Shift) to take out enemies, given they have ammo.	1. Movement Bug fix 2. Ammo Collectible 3. Shooting
Marc Edwards	1: Player Movement 2: Projectile Enemy 3: Made A Level	1: Player Movement 2: Projectile Enemy 3: Level
Cayden Jones	1. Title screen 2. Credits Screen 3. Win Screen 4. Loose screen 5. In-Game UI 6. Scoring System 7. SFX 8. Win Condition	1. In-Game UI 2. Screens 3. Scoring System

Game evidence



There is also a video that can be found in the documentation folder in the repository

Individual reflections

Jacob Ford:

I unfortunately only had a lot of time to work on this project early into the time period, and not later. As such, I was not able to help with a lot of the overarching “bring everything together” process. I did, however, get to help continue a lot of the individual elements. For example, creating a basic healing prefab object so all that was needed to use it was to drag it into the scene. In addition, a working fall zone trigger to kill the enemy should it fall off of the designated map. And most of all, I was able to create a basic enemy class script, so that all that was needed was to include the script to an enemy to provide enemy functionality, so long as movement was supplied by their own separate script, which I demonstrated by making a simple enemy to use said script.

Working with the enemy required a lot of collaboration as far as pulling and pushing changes goes, specifically with Marc Edwards, who developed the player mechanics. Because I had to make sure to create the proper functions for his player to call to affect my enemies and call the proper functions to have my enemy affect his player, we had a few times were we just we pushing and pulling small changes to work out some initial bugs that existed as we worked to seamlessly share functionality with one another till the basic enemy script was completed.

I have never had much experience with Github, so it was nice to learn how to handle basic collision errors, as well as learning how to set up the file structure and properly create and merge branches to avoid said collisions. In addition, I had never used Github issues before, so it

was nice to have a chance to explore learning how to update and close those issues with each commit (as demonstrated by some of my smaller commits in the commit log).

Jack Miller:

I started my work in a later phase of the project, so a lot of work was already in main and put together. So, I had to understand the repository before I added anything new. I decided that a good way to do this was by fixing a bug because I would have to read through the codebase and use what is already there. There was a github issue related to movement and that is where my first commit comes in. I made it so that the directional input and final direction movement were separate to allow the player to continue moving in a direction after coming in contact with a wall while airborne. Once I gained familiarity with the code base, I decided I would add a new mechanic: Shooting. To do this I first created an ammo collectible and then added the actual shooting/bullet functionality.

As for working with teammates' changes, I had to read through and edit Marc's movement script to fix that bug. I also used one of Jacob's enemy class methods, specifically killEnemy(), when adding the functionality for shooting and bullets. I learned how to use GitHub Desktop during the course of this project. Before, I would use one of the GitHub VSCode extensions or the command line and I kind of had a vague idea of what I was doing. Working with GitHub Desktop on this project allowed me to be more confident and better understand what I was doing relating to source control because of its easy GUI.

Marc Edwards:

I primarily worked on my commits during the start of the project. I created the player's movement mechanics and added jumping on enemies to defeat them, using the movement mechanics from my personal 2D game as a base. I also added an enemy that shoots projectiles at the player. To top off my commits, I also created one of the levels for the game. I worked alongside Jacob as well, primarily to get the player to defeat enemies, among other things.

I already knew how to commit things to Github before this project, but that was the extent of my knowledge. Throughout the course of this project, I learned how to create and merge branches, create and keep track of Github issues, and most importantly, I learned how to collaborate with other developers using Github.

Brendan Valleau:

To begin this project, we first started by discussing what parts we would like to focus on in the upcoming weeks. To keep it fairly simple, we decided on a simple PVE platformer game. This was pretty easily accomplished throughout the past few weeks, besides a few hiccups with the github. For the most part, we didn't have any merge conflicts. We followed your advice to each have our own separate folders, scripts, scenes, etcetera. This helped us stay focused and organized on our separate tasks. I also started out by introducing my teammates to github Issues and the kanban board, where you organize issues for a software project in one coordinated place. I talked about branches, story points, and all the other relevant information to be successful in a software development team project.

Once we got that information figured out, I created the github repository and began to initialize the project for our team. I organized the project into a folder for each team member,

and then began to locate art assets for us to use. Once that was situated, I had to painstakingly configure the tilemap to map correctly to the grid. This was extra difficult due to the pixel size of the tilemap and the layout order it auto-sliced to. I had to go in and modify the scaling of the physical tilemap itself to make the mapping more pleasant. Once that was done, I moved onto working on a basic level for the introduction to the game. I designed it to showcase the basic elements of our games, and then Marc's level would go more in depth, having a larger map to explore. And finally, I spent some time revamping the prefabs we are using in our scenes. I introduced animations for the enemies, dynamic axis flipping for the character, and scene configuration to have everyone's work fit in properly.

Cayden Jones:

For this project, I made the in game UI features such as the healthbar, timer, and score. I also made different screens for different parts of the game such as the title, win, and death screen. I also created the scoring system so that the player has a final score when they win. As I was working on this project, I would regularly fetch and pull any changes my teammates made to the code so that I was always working on the latest commit.I learned how to put issues on a board for everyone to see and work on. I also learned how to resolve merge conflicts and manuver between branches to work on code.